

# JEE MAIN REVISION MASTER: DEFINITE INTEGRATION 2026 2 QUESTIONS 3.33% WEIGHTAGE ↓33.4%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 31 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                 | Frequency | Category        |
|---------------|----------------------------------------------|-----------|-----------------|
| Very High     | Kcet Definite Integration - Core Application | 31 PYQs   | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Kcet Definite Integration - Core Application (Very High Yield — 31 PYQs)

Core Principle: Key problem pattern in Kcet Definite Integration.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int_{-6}^a (a-6)^b - 6f(x+6)dx$  is equal to..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "One of the possible functions  $f(x)$  which satisfies  $\int_{-2}^0 x^2 f(x)dx = 0$  is..." [Ans: A]
- [2025 Shift] Q#1 (MCQ): " $\int_0^1 \log\left(\frac{1}{1-x}\right)dx$  is..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

$\int_{-6}^a (a-6)^b - 6f(x+6)dx$  is equal to

- (A)  $\int_{-6}^a (a)^b f(x+6)dx$
- (B)  $\int_{-6}^a (a)^b f(x-6)dx$
- (C)  $\int_{-6}^a (a)^b f(x)dx$  [Correct]
- (D)  $-\int_{-6}^a (a)^b f(x)dx$

Correct Answer: C

**Solution Analysis:** Put  $t=x+6$  Then  $dt=dx$  Now change the limits. When  $x=a-6$ , we get  $t=(a-6)+6=a$  When  $x=-6$ , we get  $t=(-6)+6=0$  So,  $\int_{-6}^a (a-6)^b - 6f(x+6)dx = \int_a^0 (a)^b f(t)dt$  The letter used for the variable does not matter in a definite integral. So,  $\int_a^0 (a)^b f(t)dt = -\int_0^a (a)^b f(x)dx$  Hence,  $\int_{-6}^a (a-6)^b - 6f(x+6)dx = -\int_0^a (a)^b f(x)dx$  So the correct option is Option C